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Exp.No-4(A)

Title- FLOYD'S ALGORITHM (ALL PAIRS SHORTEST PATHS PROBLEM)

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In [ ]: def floyd(n, graph):
    for k in range(n):
        for i in range(n):
            for j in range(n):
                graph[i][j] = min(graph[i][j], graph[i][k] + graph[k][j])
    return graph

n = int(input("Enter the number of vertices: "))
print("Enter the adjacency matrix (enter 'inf' for infinity or '0' for no edge):")

graph = []
for i in range(n):
    row = []
    for x in input().split():
        if x.lower() == 'inf' or (x == '0' and len(row) != i):
            row.append(float('inf'))
        else:
            row.append(int(x))
    graph.append(row)

shortest_paths = floyd(n, graph)

print("=====")
print("Shortest paths between all pairs of vertices:")
for row in shortest_paths:
    print(*[x if x != float('inf') else 'inf' for x in row])
print("=====")
```

```
In [ ]: Enter the number of vertices: 3
Enter the adjacency matrix (enter 'inf' for infinity or '0' for no edge):
0 4 11
6 0 2
3 inf 0
=====
Shortest paths between all pairs of vertices:
0 4 6
5 0 2
3 7 0
=====
```